JULY-DECEMBER
2022

GOVERNMENT ENGINEERING COLLEGE PATAN

NEWSLETTER

INSTITUTE GLIMPSE

Government Engineering College, Patan (G.E.C. Patan), situated in the tranquil and scenic village of Katpur in Patan District, Gujarat, was established in 2004 with a commitment to delivering quality education in engineering and technology. Over the years, the institute has achieved remarkable growth and excellence in academics, innovation, and research. It is affiliated with the Gujarat Technological University (GTU) and functions under the administration of the Directorate of Technical Education (DTE), Government of Gujarat.

INSTITUTE VISION

"To prepare human resources with value-based competency for technical advancements and growth of society."

INSTITUTE MISSION

- **To Deliver technical programs and services to cater the current needs of society and industry.**
- **Helping industries in solving challenges by means of providing best technical human resources.**
- **To contribute in sustainable growth of society.**



ELECTRONICS AND COMMUNICATION DEPARTMENT

NEWSLETTER

ABOUT

The Electronics and Communication Engineering (ECE) Department at Government Engineering College, Patan, established in 2004, offers a four-year undergraduate program with modern infrastructure, advanced laboratories, and experienced faculty. The department provides strong foundations in electronics, communication, networking, wireless technologies, and VLSI, preparing students for careers, research, and higher education.

DEPARTMENT VISION

"To prepare engineers with essential technical knowledge, value-based professional skills for technical upgradation and societal growth."

DEPARTMENT MISSION

- **To prepare competent engineers with the ability to solve industrial problems who can raise the living standard of the society.**
- **To prepare engineers with effective communication skills, profes ional ethics and leader hip qualitie .**

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HOD'S MESSAGE



PROF. SANJAY D. JOSHI

It is my immense pleasure to present this Newsletter of the Electronics and Communication Engineering Department, Government Engineering College, Patan — a comprehensive documentation of our academic journey, industry exposure, and innovation across the years.

This edition reflects the relentless dedication of our faculty and the enthusiasm of our students. From impactful industry visits and expert lectures to project presentations and faculty development programmes, each activity represents our collective commitment to bridging academia and industry. I express my heartfelt gratitude to all stakeholders — our students, faculty, industry partners, and the college administration — who have continually contributed to making our department a centre of excellence.

~

Prof. Sanjay D. Joshi

HOD EC Department

Government Engineering College, Patan

ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- Accumulate sound technical knowledge that enables to pursue higher education and research.
- Work in multidisciplinary areas with a strong focus on innovation and entrepreneurship.
- To inculcate effective communication skills, managerial skills, as well as professional ethics for successful professional career.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- An ability to apply the conceptual knowledge of Microelectronics, Signal Processing, Microprocessors/Microcontrollers and Communication Systems to solve industrial problems.
- An ability to solve Electronics and communication Engineering problems using the latest hardware and software.

Program Outcomes (POs)

- **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- **PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).
- **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).
- **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
- PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

Program Outcomes (POs)

- **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
- **PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **PO11: Life-Long Learning:** Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

ELECTRONICS AND COMMUNICATION ENGINEERING

Faculty Members



PROF. SANJAY D. JOSHI
DESIGNATION: ASSOCIATE PROFESSOR
EMBEDED SYSTEM, IMAGE PROCESSING, MICROCONTROLLER AND MICOPROCESSOR



DR. HARESHKUMAR JUDAL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DIGITAL COMMUNICATION,WIRELESS COMMUNICATION AND DSP



PROF. MIHIRKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DSP, COMMUNICATION SYSTEMS



DR. MAULIKKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: MICROWAVE, ANTENNA, ELECTRONICS



PROF. JAYESHKUMAR C. PRAJAPATI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: COMMUNICATION ENGG, MICROWAVE ENGG, OPTICAL FIBER COMMUNICATION



PROF.SHYAMKUMAR PANKHANIYA
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEM DESIGN, VLSI TECHNOLOGY



PROF. MEHULKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: VLSI DESIGN



PROF. ROSHNIBEN CHAUDHARI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: ANTENNA, RF & MICROWAVE ENGINEERING, WIRELESS COMMUNICATION, BASIC ELECTRONICS



PROF. ANJU VASDEWANI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEMS, DIGITAL CIRCUITS AND DESIGNING, MICROPROCESSOR, MICRO CONTROLLER & INTERFACING, PROGRAMMING USING C, BASIC ELECTRONICS

FACILITIES OF EC DEPARTMENT

L201 ANALOG & DIGITAL ELECTRONICS LABORATORY

This laboratory is equipped with Analog and Digital Electronics trainers, viz.,amplifier, common emitter configuration, common collector configuration,oscillators, multivibrators, power supply, trainers using 555 IC, digital gates, flip flops, counters, and registers kits, etc. Following are few kits:

- Active band pass filter
- Automatic gain control kit
- Butterworth filter trainer
- CE configuration of transistor
- DC motor speed control using SCR



L202 SIMULATION LABORATORY

This laboratory is equipped with the latest computers with advanced simulation software, viz., MATLAB, Multisim, Visual Studio Code, Keil uVision, Atmel Studio and Xilinx etc. Laboratory is equipped with:

- DIGITAL OSCILLOSCOPE DSOX 2024A
- Workbench Electronics



L203 ELECTRONIC SYSTEM DESIGN LABORATORY

This laboratory consists of multiple electronics workbenches, having a digital storage oscilloscope, multi meter, function generator, and power supply, which can be helpful in testing and measurements of various electronics circuits or systems.



FACILITIES OF EC DEPARTMENT

L211 COMMUNICATION LABORATORY

This laboratory is equipped with Analog and Digital communication trainers, viz., AM, FM, and PM modulation demodulation, superheterodyne receiver, PWM, PPM, PAM, ASK, FSK, PSK kits, Spectrum Analyzer as well as RF and Microwave trainers for microwave communication along with active and passive components. Few trainers are as:

- Amplitude Modulation/Demodulation Trainer
- DSB/SSB AM Receiver
- Fiber Optics Demonstrator
- FM Transmitter Receiver Trainer
- Microwave Bench



L212 PROJECT LABORATORY

This laboratory is designed to provide students with hands-on experience in research and practical applications across various fields of Electronics and Communication Engineering.



L213 SIGNAL PROCESSING & VLSI LABORATORY

This laboratory consists of various trainers, viz., DSP, FPGA kits, and software required to perform operations on the trainers for Digital Signal Processing, Image Processing, Speech Signal Processing,
and VLSI experiments.

- DSP Trainer Kit
- Desktop Computer
  (ACER- AMD processor 12th generation)
- Programmable Function Generator, 1 Channel, 50Mhz (AFG3051C)



ELECTRONICS AND COMMUNICATION ENGINEERING

Faculty Activities

Faculty Achievement: NPTEL Online Course Certifications through SWAYAM Portal

Dr. Haresh L. Judal – NPTEL Course on Internet of Things (IoT)

- Dr. Haresh L. Judal successfully completed the NPTEL Online Course on Internet of Things (IoT) through the SWAYAM Portal from 25 July 2022 to 14 October 2022. The course provided comprehensive knowledge of IoT architecture, embedded systems, sensors and actuators, communication protocols, cloud computing, security, and various industrial applications. This certification enhanced his understanding of emerging IoT technologies and enabled him to incorporate practical, industry-oriented concepts into teaching, research, and student project guidance, thereby contributing to improved learning outcomes.

Prof. Mehul L. Patel – NPTEL Course on Fundamentals of Artificial Intelligence

- Prof. Mehul L. Patel successfully completed the NPTEL Online Course on Fundamentals of Artificial Intelligence through the SWAYAM Portal from 25 July 2022 to 14 October 2022. The course covered the core principles of Artificial Intelligence, including intelligent agents, problem-solving techniques, knowledge representation, search algorithms, and the fundamentals of machine learning. The certification strengthened his expertise in AI and supported the integration of modern AI concepts into classroom teaching, research activities, and student learning, promoting innovation and technological advancement within the department.

World Paper Bag Day on 12th of July 2022



World Paper Bag Day is celebrated every year on 12th July. The main motive behind this day is to spread a positive message of using paper bags and reducing the use of plastic bags. This year?s theme was If you are fantastic, do something dramatic to cut plastic, use paper bags. In order to spread awareness NSS GEC Patan celebrated World Paper Bag Day on 12th July in our college. An online competition was organized by NSS in which participants were asked to make paper bags and submit it offline. In which many NSS volunteers and college students participated. Many participants were also from other states. The best paper bags were chosen out of all responses by Prof. Roshni Chaudary (E.C. Department), Prof. Payal Prajapati (CSE Department), and Prof. Priyanka Patel (Mechanical department).

The top three winners were awarded with Bird's Nest to deliver positive message of saving and caring of birds. All the participants of the event were appreciated by providing E-Certificates.

The NSS Unit of Government Engineering College, Patan celebrated World Paper Bag Day on 12th July to promote environmental sustainability and encourage the use of eco-friendly alternatives to plastic bags. An online paper bag making competition was organized, attracting enthusiastic participation from NSS volunteers and students. The best entries were selected by a panel of faculty judges, and the winners were awarded Bird's Nests to spread the message of bird conservation. E-certificates were presented to all participants, while NSS volunteers distributed handmade paper bags among the faculty members to raise awareness about reducing plastic usage.



GURU PURNIMA on 13th of July, 2022



On 13th July 2022, NSS Volunteers celebrated Guru Purnima by doing Tilak to every professor. Volunteers also invited them for the celebration of this event. The celebration of Guru Purnima was held in electrical seminar hall, GEC Patan. The seminar hall was beautifully decorated by NSS volunteers. At the beginning of the event, Saraswati Vandana and deep Prajvalan took place. After that, the classical dance on the importance of the guru was performed by Dhruvanshi. After completion of the dance, everyone stood up at their place and NSS song was sung by our Volunteers.

Teaching might even be the greatest form of arts since the medium is the human mind and spirit

Guru Purnima is a tradition dedicated to all the spiritual and academic gurus, who are evolved or enlightened humans, ready to share their wisdom, based on karma yoga. The Guru Disciple tradition is a unique feature of Bhartiya culture. The meaning of the guruis Gu means Darkness and Ru means removal. The Guru is the one who removes ignorance from our life.



Rakhi Making Competition on 20th July, 2022

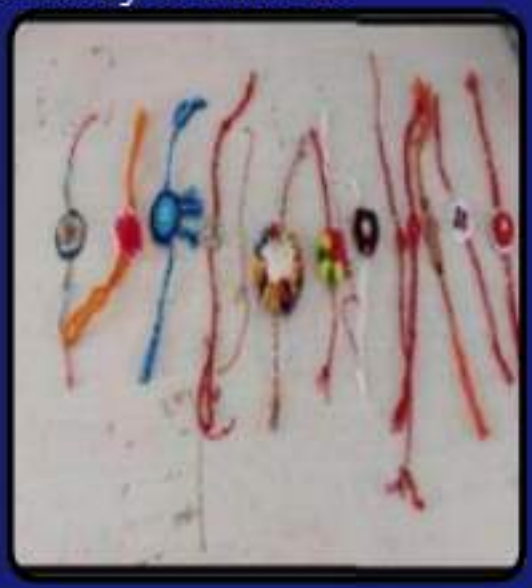
Raksha Bandhan is special hindu festival which is celebrated in India and countries like Nepal to symbolize the love between brother and sister. The occasion of Raksha Bandhan is celebrated on the full moon day of the hindu calendar in the month of shravana (July/August). On this day, sisters tie rakhi on the wrists of their brother to protect them against evil influences ,and pray for their long life and happiness and in return as a gift brothers gives promises that they will protect her from any harm and keep her always happy. This does not means that the brother protects the sisters . Our soldiers also protect us at the border. They stay away from their home to protect us . So for this, NSS GEC Patan with collaboration of Yuva Foundation organized an event on the occasion of In this way, the students came to knew that their innovative ideas can get the nice place to be presented, and they get encouragement to do Startup and give their contribution to increase the economic of the India.

Major Outcomes of the Event

➢ The people of yuva foundation gave the description about there foundation.

- These rakhi will be sent to armed forces on borders who are living away from homes.
- Through RAKHI MAKING competition, itshow the creativity of students.





Voting awareness program on 28th of July, 2022

Bad government are elected by good citizens who don't vote.

Voting is the expression of our commitment to ourselves,one another,our country and this world.The law does not require citizens to vote, but voting is a very important part of any democracy. By voting, citizens are participating in the democratic process. Citizens vote for leaders to represent them and their ideas, and the leaders support the citizens' interests.everyone in the country irrespective of the caste, colour, gender, creed and income earned is allowed to vote for his own representative. On 28th July 2022, NSS Volunteers celebrated World Nature Conservation Day.On this day Volunteers do a Voting awareness program under the Electoral Literacy Club. The event was organized under the guidence of Dharmesh sir. The celebration of this event was held in civil seminar hall, GEC Patan. The seminar hall was beautifully decorated by NSS volunteers. At the beginning of the event, Saraswati Vandana and deep Prajvalan took place. After that, everyone stood up at their place and NSS song was sung by our Volunteers. Then speech on voting awareness was given by Anika Rehan. Also, volunteers performed drama on importance of voting.

Major Outcomes of event

- **We learn how to work in team in such huge event.**
- **Importance of Voting as a responsible citizen of country.**
- **We learn about voting procedure**





Gandhi Jayanti on 2 nd of October, 2022

On the occasion of 153rd birth anniversary of Late Shri Mohandas Karamchand Gandhi popularly known as Mahatma Gandhi The Father of Our Nation NSS Unit of GEC Patan organized an online event for celebrating the same. The Event was started by prayer song by NSS Volunteer Smruti Lakdawala. Then NSS song was played. An online drawing competition was organized whose responses were asked to submit through Google Forms. Participants were given 48 hours to submit their responses, as a result of which many numbers of participants had submitted their responses. Out of which 3 Winners were selected by our judges of the event Dr. A.K.Chaudary Sir and Prof. Payal Prajapati Mam. The winners of Drawing Competition were as mentioned below.Speech Competition was held in which Participants were asked to give speech on topic Role of Mahatma Gandhi in Independence of India. The Winners of Speech Competition were appreciated with book and all enthusiastic participants get E-certificate. And Dr. H.S.Bhutadiya sir gave a great speech on personality of Gandhiji. At the end of Event Program Officer of NSS GEC Patan Proff. N.K.Dabhi thanked the guests for their valuable presence and also thanked audience for being present.

Major Outcomes of Event

• It was great experience for all to become witness together of such a historic moment. • Proud moment for all Indians and cheers for our ISRO scientists.

• From these watch together students acknowledge about upcoming launches and projects of ISRO.







THALASSEMIA CAMP on 10th October, 2022

A Thalassemia camp was organized at the Government Engineering College, Patan on Monday, 10 October 2022 for the thalassemia test of all the students. The camp was arranged at the workshop building on campus. It was organized by the Indian red cross society, Ahmedabad. The camp began at 10:30 am. All the students' samples were collected for the thalassemia test during this camp.

NSS volunteers helped the students for filling up their acknowledgment forms for the camp. There were a large number of students were present during the camp for getting their blood tested. About 200 students from various branches and semesters provided their blood samples for testing. Time slots were arranged for students according to their department for the ease of collection of blood samples. The medical team administered to collect the blood samples from thestudents. Awareness was created among the students regarding thalassemia, its types and the effects of this blood disorder in our life. The importance of thalassemia testing was also elaborated.



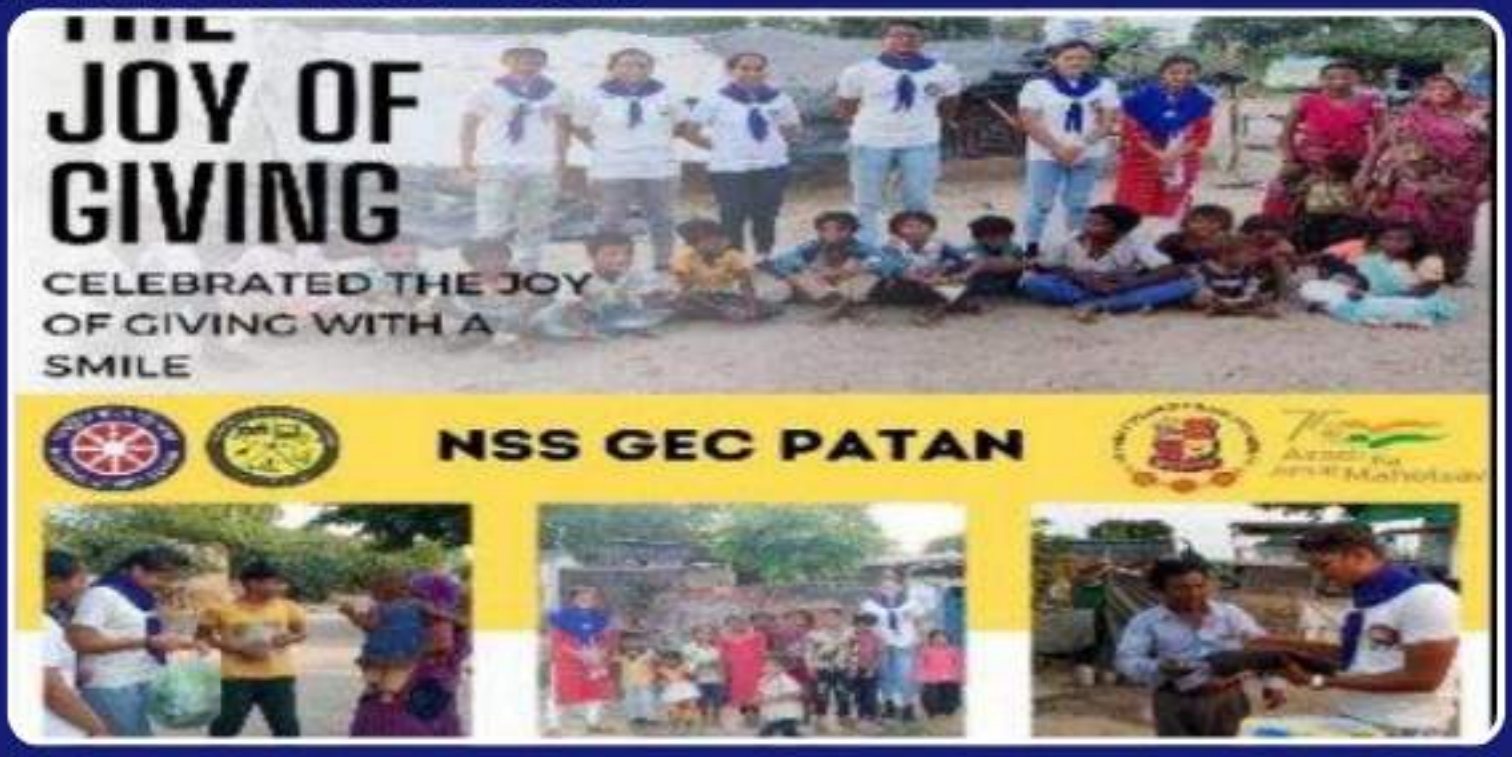


Major Outcomes of Event

- Spreading awareness among the students about thalassemia and early diagnosis.
- Stress on the importance of a healthy diet and regular exercise for thalassemia patients.
- Encourage the students to donate blood regularly.

JOY OF GIVING on 20th October, 2022

The joy of giving is organized by NSS volunteers. In Patan city on Thursday, 20th October 2022 to give poor people clothes and sweets. The time of doing this work is 20th Oct 3:00 pm to 7:00 pm. This event is organized to give poor people clothes and sweets the way NSS volunteers collected all wearable clothes from the neighborhood and divide them into the category after that NSS volunteers organized some sweets. After organizing all things NSS volunteer leave for distribution works in the south side of Patan city. First of all, we went to a place which was close to our homes and there we gave the children clothes to wear and gave them some sweets. After that, we went to the nursery and gave sweets to the all workers. Then we went to where the poor people lived and gave clothes and sweets and after that, all the children gathered there and gave sweets and chocolates after that we came back.



Celebration of Indian Constitution day on 24th November, 2022

Indian Constitution Day is celebrated on 26 November. The celebration of Indian Constitution Day has been started in the year 2015. From 26 November 2015, we are celebrating this day.

On the occasion of Indian Constitution Day, the event had been organized by the volunteer of the NSS unit. At the beginning of the event, the Deep Pragatya ceremony has been held in the presence of many professors and respected principal sir. Then after welcoming all the respected members had done.

The speech was given by NSS Volunteer Vikhnesh Pillai. After that expert lecture had been taken by Prof. Sohan G Prajapati about the importance of Indian Constitution Day which gave an overview of it. The event ended with the pledge and National Anthem.

Major Outcomes of The Event

- Major Outcomes of Event
- Got information about the Indian Constitution.
- Everyone took a pledge to secure Indian Constitution.
- We were glad to know about the amazing facts about our constitution.
- Many important articles were explained in detail.





Event on Har Ghar Dhyan on 29thDecember, 2022







Har Ghar Dhyan is an event that falls within Azadi ka Amrit Mahotsav. Har ghar dhyan is an event to motivate everyone towards meditation. Meditation can give you a sense of calm, peace and balance that can benefit both your emotional well-being and your overall health. You can also use it to relax and cope with stress by refocusing your attention on something calming. Meditation can help you learn to stay centered and keep inner peace. So for this NSS Unit of GEC Patan organized an event Har Ghar Dhyan= on 29th December, 2022. NSS volunteer Anjali Thakkar hosted this event. She gave some information on meditation and conducted meditation for 15 minutes. Many students and Faculties attended this event and everyone participated in meditation. Everyone felt freshand a lot of peace after this meditation.

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